### *Solution*

The slope of *M-H* graph is a measure of the magnetic susceptibility, which is given by

$$\chi_m = \frac{M}{H}$$

Material X: Slope is positive and larger value. So, it is a ferromagnetic material.

Material Y: Slope is positive and lesser value than X. So, it could be a paramagnetic material.

Material Z: Slope is negative and hence, it is a diamagnetic material.

## 3.7 MAGNETIC EFFECTS OF CURRENT

### 3.7.1 Oersted experiment

In 1820 Hans Christian Oersted, while preparing for his lecture in physics, noticed that electric current passing through a wire deflects the nearby magnetic needle in the compass. By proper investigation, he observed that the deflection of magnetic needle is due to the change in magnetic field produced around current carrying conductor (Figure 3.25). When the direction of current is reversed, the magnetic needle is deflected in the opposite direction. This lead to the development of the theory 'electromagnetism' which unifies two branches in physics namely, electricity and magnetism.

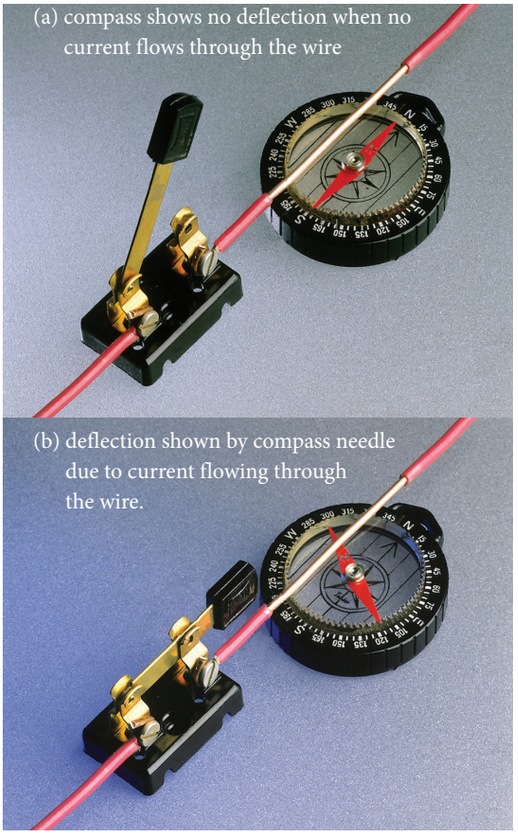



**Figure 3.25** Oersted's experiment - current carrying wire and deflection of magnetic needle

### 3.7.2 Magnetic field around a straight current-carrying conductor and circular loop

**(a) Current carrying straight conductor:**

Suppose we keep a magnetic compass near a current-carrying straight conductor, then the needle of the magnetic compass experiences a torque and deflects to align in the direction of the magnetic field at that point. Tracing out the direction shown by magnetic needle , we can draw the magnetic field lines at a distance. For a straight current-carrying conductor, the nature of magnetic field is like concentric circles having their common centre on the axis of the conductor as shown in Figure 3.26 (a).



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